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RESEARCH ARTICLE

Mineral Extraction Scenarios From an Asteroid via Optimization Considering Rotational Dynamics

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ABSTRACT This study examines how removing surface materials affects the rotational dynamics of an asteroid to design efficient asteroid mining plans. The simulations solve Euler's equations and track variations in the inertia tensor, angular momentum, and energy throughout the removal process. Two distinct methods (random and optimized) are employed to remove 30% and 70% of the total asteroid mass. Findings demonstrate that random removal results in large deviations from the initial angular velocity, most noticeably when a greater percentage of the asteroid's mass is removed. It is also shown that random removal leads to irregular shapes as they gain angular velocity with higher percentages of mass removal. The optimized removal strategy achieves substantially lower dynamical error even when large portions of the surface are extracted and suggests excavation sites should be symmetrical, especially along the z-axis, while avoiding excessive polar areas, to preserve rotational stability. This paper shows the effectiveness of targeted material removal in maintaining dynamical stability and structural integrity and offers practical guidance for the safe and efficient design of future asteroid mining missions.

INDEX TERMS Asteroid engineering, asteroid mining, material removal, rotational dynamics, optimization.

I. INTRODUCTION

The increasing global demand for natural resources and the fast depletion of terrestrial reserves have necessitated the exploration of extraterrestrial sources as a viable substitute [1], [2]. Near-Earth asteroids (NEAs) are attracting more attention as they are close to Earth and possibly contain precious materials, i.e., metals and volatiles [3]. The success of asteroid mining technologies would, thus, revolutionize industries, provide resources for budding space exploration, and help to lay the foundations for sustainable off-Earth economies [4], [5].

However, resource extraction from asteroids is enormously challenging; one of the technical challenges is maintaining the rotational stability of asteroids before, during, and after mining operations. Changes in an asteroid's angular velocity or rotational state due to material removal can complicate subsequent operations, including the navigation of equipment, resource transferring, future mission designing, and

possible redirection of the asteroid, which might lead to an impact by Earth [6], [7]. This challenge highlights the vital need for optimization strategies to minimize any appreciable and detrimental changes in the asteroid's rotational dynamics during the mining process.

In this paper, we address the problem of optimizing material extraction from asteroids to minimize deviations in their rotational characteristics. Particle Swarm Optimization (PSO), a widely used optimization technique, was utilized in developing the framework that identifies excavation strategies to maintain the dynamic stability of the asteroid. Dynamic simulations then provide insight into the effects of the excavation location and the asteroid's rotational state, allowing for practical points of reference to minimize changes in angular velocity. Hence, the findings will help provide the very basis for building efficient and indeed sustainable asteroid mining methodologies.

In recent decades, asteroid mining has become a favored topic for discussion in many areas as a possible solution to the increasing demand for critical resources, coupled with the limitations of terrestrial mining [8], [9]. Early studies

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on resource utilization from asteroids, mainly by Sonter [10] and Ross [11], underscored the economic and technical feasibility of extracting materials from NEAs. These works laid the groundwork for ascertaining NEAs as a viable source of metals, volatiles, and other precious materials. Subsequent research built on these initial works, focusing on the search for appropriate candidates, mining technologies, and further operational challenges associated with these areas of study [12], [13].

On the other hand, many precarious challenges arise while framing optimum scenarios in asteroid mining. Some of them include, but are not limited to, the dynamic stability of the asteroids during resource extraction. Scheeres Schweickart [14] specifically explored the rotational dynamics of small bodies and came to demonstrate the fact that their rotational states are quite sensitive to external perturbations. What drops further from their conclusion would signify the importance of having some manner of rotational stability in the very successful proceedings of asteroid mining operations. Sanchez and Scheeres [15] find further utility in analyzing also the structural and rotational behavior of rubble-pile asteroids, consisting of a large volume of NEA, meaning that they talk about excavation-related risks imposed by perturbations.

The comprehensive investigation performed by Toshihiko Kadono et al. [16] sheds worthwhile illumination on the rotational characteristics related to asteroids, specifically referencing those that experience fast rotation. The authors point out that the resulting fragments of impact events generally have spin rates in proximity to the critical limit imposed by material strength and cohesive forces. This work is valuable in determining the structural integrity of an asteroid under an external perturbation, which is important in providing a design background for safe mining operations. On the other hand, the research conducted by Brack and McMahon [17] deals with one or the other rotational dynamics under the influence of excavation processes. discusses the effects of removing material from an asteroid on the asteroid's rotation and stability. The paper describes the mass and inertia of the asteroid as it changes in a stepwise mode as material is removed and describes how the changes in mass and inertia impact the angular velocity of the asteroid. The study demonstrated that even without additional forces, mass removal can significantly impact rotation, particularly at higher levels of removal, underscoring the importance of considering these effects in asteroid mining mission design. Thus, these findings pinpoint the urgent need to consider rotational dynamics in asteroid mining methodologies and are closely related to the principal objectives of this research. Xie et al. [18] provided a framework that evaluates NEAs as potential mining targets, based on their orbital properties, mineral composition, and accessibility for long-term missions, highlighting the necessity for asteroids to have fairly easy trajectories and high resource potential if a mission is to be feasible and economically productive. It further has significance for proposing strategies to prioritize NEAs due to mining

operations. On the other hand, Lauretta et al. [19] investigated the dynamics of particle ejection processes from the surface of asteroids due to external forces, such as solar radiation or impact events. They note that particle ejection can greatly modify the surface structure, thereby potentially destabilize the asteroid's spin state and complicating all mining operations. The two studies together emphasize, among others, the importance of target selection and the comprehension of surface behavior, both of which play a role in the successful and sustainable Plan for Mining Outer Space.

Despite these advancements, the research on the integration of rotational dynamics into the optimization of asteroid mining operations remains quite limited. Most of the existing studies aimed to maximize resource yield or minimize the risk of excavation. Some works focused on the control and detumbling of asteroids before the mining process. The control and detumbling of asteroids are crucial operations for both mining and scientific missions, as many small bodies exhibit significant rotational motion or irregular tumbling, which pose substantial challenges to spacecraft operations. Reliable detumbling ensures that landers, mining equipment, or redirect missions can safely interact with and manipulate an asteroid. So, effective control and detumbling enable the asteroid to be maneuvered to a safe and predictable rotational state [20], [21], [22]. In this context, little attention has been paid to retaining the original rotational characteristics of the asteroid. Another aspect of non-controlled extractions from an asteroid which is worth noticing is about binary asteroids. One of the key mechanisms for the formation of binary asteroids, particularly among rubble-pile asteroids, is mass shedding caused by an increase in rotational speed of the parent body. During mining operations, sudden and irregular increases in angular velocity due to resource extraction can similarly induce structural failure or fragmentation of the asteroid. This not only risks destroying the asteroid but can also halt the entire mining process, presenting a significant operational challenge. Therefore, controlling rotational dynamics is critical to maintaining the structural integrity of rubble-pile asteroids during surface material removal [23], [24], [25]. So, changes in angular velocity or rotational state will consequently affect all mining operations, spacecraft trajectory, and possibly later redirection efforts on asteroids. This paper continues this body of work by looking at the interaction of rotational dynamics and resource extraction during asteroid mining. In this work, we aim to introduce an optimized excavation scenario by using particle swarm, which will, in turn, minimize the deviation from its original rotational behavior in consideration of better sustainability for mining.

II. NEAR-EARTH ASTEROID

NEAs are very much on the priority list due to their proximity to Earth and their possibility of being sources of useful resources to aid in the exploration of space and its terrestrial applications. Made of basic materials: clay, silicate rocks, nickel, iron, and water, these NEAs offer a strong need for

extraction [26], [27], [28], [29], [30]. For classification, the asteroids are generally divided into: C-type- carbonaceous, which represent more than 75% of known asteroids. They are water-rich, volatile-rich, and carbonaceous-rich. S-type siliceous asteroids, which account for about 17% of known asteroids, are rich in silicate minerals, nickel, and iron, making them useful in industrial applications. M-type (metallic) or X-type are comparatively less in number but are equally valuable. They are made of nickel, iron, and rare metals like platinum, cobalt, and gold [31].

NEAs are significant not only for their accessibility and resource potential but also because some of them pose a potential threat to Earth. They include primitive carbonaceous asteroids that are among the most reachable and some of the most hazardous objects identified. To better understand NEAs, Milani, et al. conducted a dynamic classification by numerically integrating the orbits of 410 known NEAs over the years [32]. Based on their findings, NEAs were divided into six orbital classes, each named after a well-known member: Geographos class, Toro class, Kozai class, Alinda class, Eros class, and Oliato class. In addition to these, there are comet-like objects that resemble asteroids in appearance but follow cometary orbits.

Studies concerning the rotational dynamics of NEAs have also proven critical in revealing clues to their structure and behavior. Hergenrother and Whiteley [33] stated that those NEAs with rotation periods of less than 2 hours may have an internally strong structure, while the others with longer periods form clusters that are usually held together by gravity. Furthermore, at least two-thirds of NEAs with an absolute magnitude greater than 20 exhibit rapid rotation, with periods significantly shorter than 2 hours.

FIGURE 1 shows the total count of identified Near-Earth Asteroids (NEAs) over time.¹ It includes totals for NEAs of various sizes, including those larger than approximately 140m and those larger than approximately 1km. FIGURE 2 depicts the cumulative count of identified Near-Earth

¹<https://cneos.jpl.nasa.gov/stats/size.html>

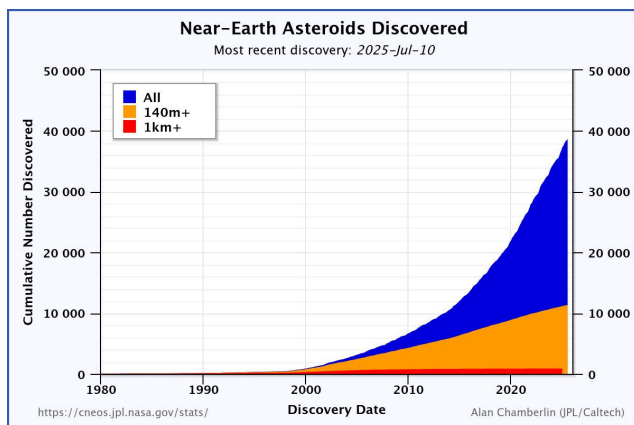


FIGURE 1. The cumulative number of known NEAs as of July 10, 2025.

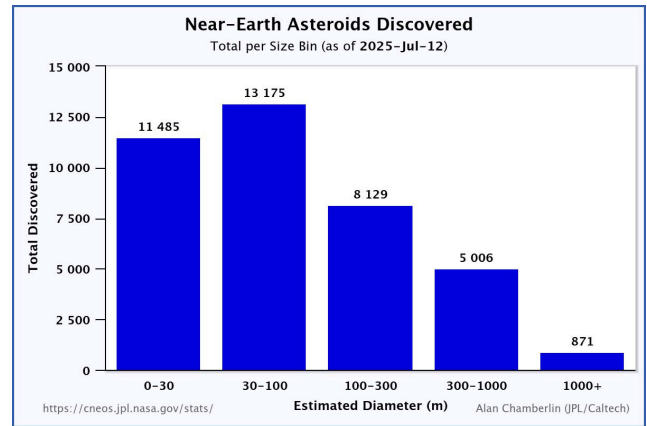


FIGURE 2. The total number of known NEAs grouped according to their estimated sizes as of July 12, 2025.

Asteroids (NEAs) categorized by their estimated sizes.² The smallest category includes NEAs less than ~30m in diameter, while the largest category includes NEAs greater than ~1km in diameter.

III. ROTATIONAL DYNAMICS

In this paper, we modelled Bennu in such a way that it reproduced an irregular shape and mass distribution using a cube-based approach. The high-resolution 3D model of Bennu that was obtained from the NASA OSIRIS-REx mission was constructed by a grid of small cubes, each representing a discrete volume element of the asteroid.³ Its center of mass was pointed out at the origin of the coordinate system to align the model correctly. The cube grid was visualized with PyVista (an open-source package for Python), and coordinates were scaled to kilometers. The cube-based model thus served as the foundation for the inertia tensor calculation and rotational dynamics simulation based on Euler equations. This method provides a reliable framework for studying irregular-shaped celestial bodies' properties and dynamics. FIGURE 3 depicts the 3D model of asteroid Bennu.

TABLE 1 shows the physical and orbital properties of asteroid Bennu [34].

It is worth noticing that while Bennu is treated as a rigid body for modelling purposes, scaling arguments show that its centrifugal stresses under residual spin-up (~10–20 Pa) are of the same order as the cohesive stresses expected from van der Waals forces in rubble-pile regolith (~1–25 Pa) [15]. This implies that cohesion can shift the critical spin limit by tens of percent, as shown in detailed finite-element analyses of other rapid rotators such as (29075) 1950 DA, where a cohesion level of 75–85 Pa in the core was required to prevent failure at a 2.1 h period [35]. In contrast, more slowly rotating M- and C-type NEAs ($P > 6$ h) have dimensionless spin parameters $W < 0.2$, suggesting that centrifugal stresses are well below realistic cohesion levels and thus their dynamical

²<https://cneos.jpl.nasa.gov/stats/totals.html>

³<https://svs.gsfc.nasa.gov/5069>

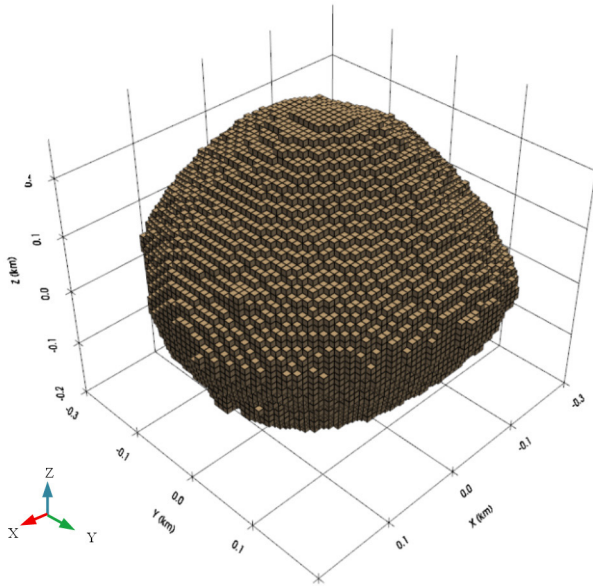


FIGURE 3. Asteroid Benu 3D model.

TABLE 1. Benu's physical and orbital characteristics.

Parameter	Value
Density[g/cm ³]	1.26
Principle diameter[km], DX	0.565
Principle diameter[km], DY	0.535
Principle diameter[km], DZ	0.508
Aphelion [AU]	1.355
Perihelion [AU]	0.896
Semi-major axis [AU]	1.126
Orbital period [day]	436.648
Mean anomaly [deg]	101.703
Inclination [deg]	6.034
Eccentricity	0.203

behavior could be more conservative than Benu's. Mapping scenarios in this non-dimensional parameter space, together with inertia ratios, provides a way to judge the transferability of Benu-based results across the broader NEA population. It is even consistent with the general use of non-dimensional dynamic metrics used in spacecraft and small-body studies [36].

FIGURE 4 illustrates the inertial frame (N) and a body-fixed frame (B) used to describe the motion of an asteroid; the inertial frame, defined by axes X, Y, and Z, provides a fixed reference, while the body-fixed frame, defined by axes x, y, and z, has its origin at the asteroid's center of mass and rotates with it. The vector R represents the position vector of the asteroid's center of mass in the inertial frame, and the angular velocity ω indicates the asteroid's rotation within the body-fixed frame.

We assume that the asteroid is constructed of small cubes. So, the total inertia moment of the asteroid can be calculated

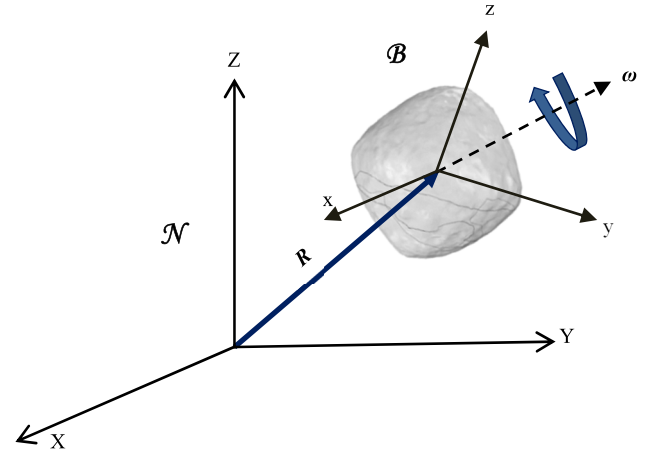


FIGURE 4. Reference frame and coordinate system.

as the summation of the elements:

$$\begin{aligned}
 I_{BT,x} &= \sum_{i=1}^n \frac{(ma^2)_{x,i}}{6} + m_i (y_i^2 + z_i^2) \\
 I_{BT,y} &= \sum_{i=1}^n \frac{(ma^2)_{y,i}}{6} + m_i (x_i^2 + z_i^2) \\
 I_{BT,z} &= \sum_{i=1}^n \frac{(ma^2)_{z,i}}{6} + m_i (x_i^2 + y_i^2)
 \end{aligned} \quad (1)$$

where, $I_{BT,x}$, $I_{BT,y}$, $I_{BT,z}$ are the total inertia tensor of asteroid Benu to the three major axes and $\frac{(ma^2)_{x,i}}{6}$, $\frac{(ma^2)_{y,i}}{6}$, and $\frac{(ma^2)_{z,i}}{6}$ are the inertia moment of each element in where m is the mass of each element and a is the length of each edge, measuring from its center of mass. m_i is each particle's mass, which is multiplied by the distance between the asteroid's center of mass and each particle's center of mass, considering their axis. Benu is considered a rigid body in which cubes can be removed. The rotational dynamics of the rigid body are described using the Euler equations. The general form of the Euler equation in the fixed-body frame is:

$$\frac{d}{dt} (I_{BT}(t)\omega) + \omega \times I_{BT}(t)\omega = M \quad (2)$$

in which $I_{BT}(t)$ represents the inertia tensor changing over time due to the extraction of minerals, ω is the angular velocity, and M is an external torque. The effects of spacecraft maneuvers, e.g., landing and takeoff, and excavation processes, e.g., particle ejection, thruster interactions, can be modelled as external torques and incorporated to make the model more accurate. Nonetheless, in this paper, we assume that no torque is applied to the problem. This time-changing inertia tensor is calculated after each element removal as a discrete quantity and afterward inserted into the Euler equation. By solving the system of differential Eq. (2), the change of angular velocity over time can be obtained.

IV. OPTIMIZATION PROBLEM

Our primary objective is to minimize the changes of angular velocity as the key metric, which changes due to the material removal, leading to an alteration in the moment of inertia. Uncontrolled changes in rotational velocity can lead to unpredictable asteroid behavior, posing risks to spacecraft missions. To mitigate these risks and preserve mission plans, it is crucial to minimize alterations to the asteroid's rotational velocity. Optimization methods were employed to address the challenges of asteroid mining. PSO, introduced originally by Kennedy and Eberhart [37], is a robust and computationally efficient optimization technique without having to incorporate too many tunable parameters and failure risks in engineering problems connected to asteroids.

In this study, we aim to extract approximately 30% and 70% of the asteroid's elements, simulating the process from its surface to its core. It is also possible to add other constraints to the optimization problem, such as economic restrictions and preserving the angular momentum vector, to minimize the dynamical behavior changes. For the first one, our goal is just to investigate the dynamical analysis, so these kinds of constraints are ignored. The second one would address the operational concern of a landed spacecraft, which should accommodate any misalignment between angular velocity and angular momentum. However, in this paper, we just examine the minimization of the changes in angular velocity magnitude. It could provide a base for which a more complex cost function can be developed in future work. We examined two cases (30 and 70%), but it is possible to increase the mass removal process in 10% increments. This would enable us to determine the maximum percentage of material that can be extracted from each asteroid without compromising its safety. We define an objective function that compares the initial angular velocity, $\hat{\omega}(i)$, which is not assumed to be zero and the magnitude of the calculated angular velocity, $\omega(i)$, obtained by solving Euler's rotational dynamics equation with the asteroid's updated inertia tensor after each step of mass removal. The difference represented as $\omega(i) - \hat{\omega}(i)$, is squared to compute the error for each particle, i^{th} , in the valuable set. The mean squared error (MSE) is then calculated by averaging the squared errors across all n data points, providing a measurement of accuracy in our predictions.

$$MSE = \frac{1}{n} \sum_{i=1}^n (\omega(i) - \hat{\omega}(i))^2 \quad (3)$$

where MSE is our objective function. To minimize an objective function, we utilize the PSO algorithm. TABLE 2 shows the PSO parameters used in this work.

Aiming for the possible complications to be dealt with beforehand, to avoid obstacles from the extraction procedure in question, is what should be investigated here. Knowing the structure of the asteroids will further enhance one to develop proper methods for extraction with the least disruption to the mission.

TABLE 2. PSO parameters.

Parameter	Value
Swarm size	50, 100, and 200
Maximum iterations	3500, 2500, 800
Cognitive factor	1.5
Social factor	1.5
Inertia weight	0.7
Random value	[0,1]

Carrying out risk assessments and subsequent mitigation controls will surely minimize other negative impacts. Mitigating the angular velocity changes and rotations will allow many propitious conditions for extraction with a retained mission integrity.

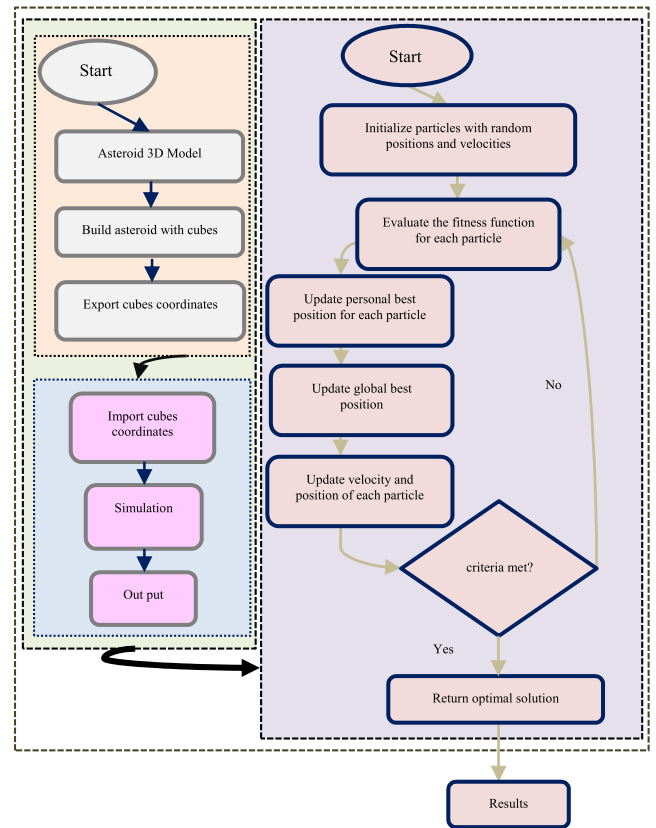


FIGURE 5. Problem-solving framework.

FIGURE 5 depicts the process done in this paper. The process started with the importing of a high-definition 3D model of the asteroid through the NASA OSIRIS-Rex mission. Then, it gets filled by creating a grid of small cubes within that 3D space, where it checks if each cube is inside the surface of the asteroid. Then, the interior cubes are assigned mass respective to the distribution in the density of the asteroid. The entire cube-based model has its center of mass calculated with a resultant change in grid alignment so that, in simulation, the center of the model aligns with the origin in space.

By creating a cube-based model, the inertia tensor of the model can be computed with the mass and position of each

cube as input. This inertia tensor is used to simulate rotational motion from Euler's equations. The angular velocity and energy are examined for the analysis of the asteroid's behavior. To conduct an optimal extraction process, the PSO algorithm identifies which cubes could best be sacrificed with the least effect on the angular velocity of the asteroid.

V. RESULTS AND DISCUSSION

The objective of this paper is to minimize rotational disturbances during asteroid mining by finding suitable and optimum locations for material extraction. This was achieved through a combination of dynamic simulations based on Euler's equations and PSO. Our algorithm is examined for two scenarios, and the obtained results show rather convincingly that PSO optimization proposes solution sets for distributed surface-level excavation, which would keep the asteroid very close to dynamic equilibrium. In scenario 1, 30% of the total asteroid mass was removed, while in the second scenario, the removal mass is 70%. Firstly, the PSO convergence and the characteristics of the optimized excavation sites are examined, followed by an analysis of their impact on the asteroid's rotational dynamics.

FIGURE 6 shows the optimization results for the fitness plot, demonstrating that both Scenario 1 (30%) and Scenario 2 (70%) achieve rapid and effective convergence for three swarm sizes, i.e., 50, 100, and 200 particles. The convergence behavior of the PSO algorithm under these three swarm sizes shows that while all configurations eventually reach near-zero fitness values, the speed of convergence varies significantly with the number of particles. For 50 particles, the best fitness stabilizes below almost 0.01 and 0.05 after approximately 2200 and 2300 iterations for scenarios 1 and 2, respectively. For 100 particles, the 0.01 value is achieved within around 1400 iterations for scenario 1 and 1700 iterations for scenario 2. For the configuration with 200 particles, the convergence is significantly faster, with the best fitness dropping below 0.01 after only approximately 40 iterations in scenarios 1 and 2. In terms of results, all three cases achieve very similar minimum fitness values, indicating that increasing particle numbers does not improve the ultimate solution quality but has a strong impact on convergence speed. Furthermore, the comparison between Scenario 1 (30%) and Scenario 2 (70%) across all cases reveals no significant difference, although Scenario 2 generally shows faster reduction in fitness during the early iterations. These results illustrate that an increase in the swarm size provides mainly an improved efficiency of the optimization process, in a reduced number of iterations to reach an acceptable solution. However, since computation costs are limited by the number of particles as well as iterations, 200 particles may take almost three times as many function evaluations per iteration as a swarm with 50 particles. Hence, to re-run the optimizer on board where processor power is limited, smaller swarms with more iterations may provide a more reasonable tradeoff between exactness and resource usage. Notably, the rapid drop in fitness value in the first iterations is

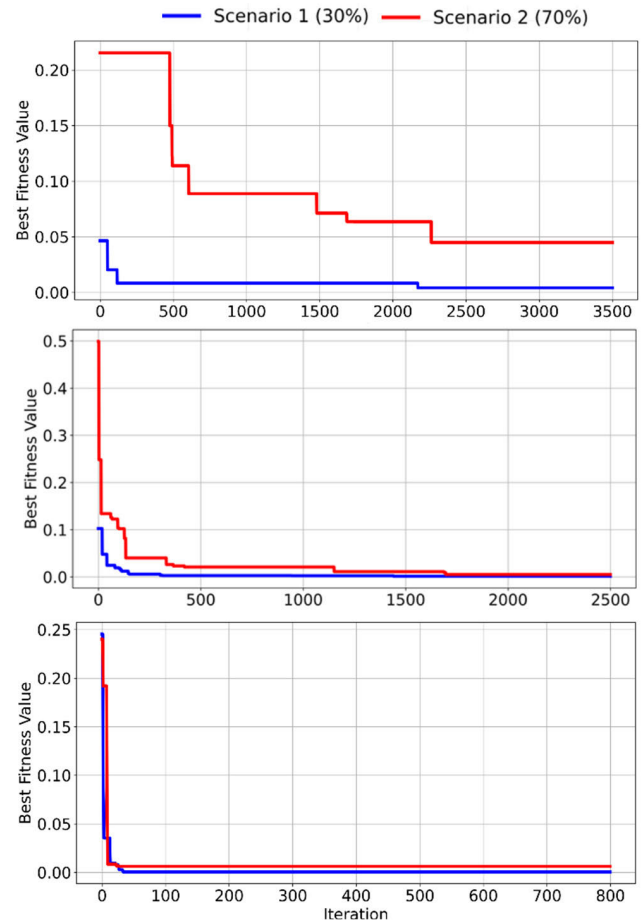


FIGURE 6. Fitness value convergence for swarm size 50 (Top), 100(middle), and 200 (bottom).

a testament to the optimization approach, while the eventual settling of fitness values at very low values generally indicates that solutions are well-resolved. Overall, the optimization results validate the PSO algorithm to optimize the removal sequence while maintaining stability during asteroid mining operations.

FIGURE 7 and FIGURE 8 compare the asteroid shape outcomes for the two scenarios, which reveal that the extent and location of material removal dramatically influence the final morphology. In the scenario with a lower removal fraction (scenario 1), the asteroid maintains a more regular and cohesive shape, with only minor surface perturbations and most of its original structure intact. The other scenario (scenario 2) with the greater removal fraction produced asteroid shapes that were rougher, more angular, larger and more pronounced excavations, and the deformed shape and geometry deviated from both the original shape and original geometric considerations and then the loss of mass, necessary for a reshaping of the details of the shape.

Notably, the optimization algorithm identifies specific locations on the asteroid's surface where material can be extracted with minimal effect on the asteroid's shape and dynamical stability. The optimizer often selects sites that are less critical in effect. It often looks to less dynamically

important surface cubes or distributed sites to minimize the alteration of shape and reduce the overall risk of rotational instability. The potential sites recommended by the optimizer represent the best possible options for the extraction of material, as they provide the greatest amount of material while maintaining operational stability and safety for the asteroid mission. Such views of optimized site areas are important and informative for the realization of asteroid mining in the future and mission planning. These sites are well disturbed along the z-axis and deviated from polar regions. Additionally, the removal sequence for both scenarios is shown in the appendix.

FIGURE 9 and FIGURE 10 illustrate the comparison of the residual angular velocity for both 30% (Scenario 1) and 70% (Scenario 2) of material removal, respectively, evaluated using random and optimized removal approaches, which demonstrate significant differences in dynamical response dependent on the extraction strategy and associated removal fraction. For the cases of random removal, the residuals for each axis of angular velocity displayed continued oscillations and were deviated from zero, where the magnitude and complexity of oscillation increased from 30% to 70% removal. This trend highlights how material extraction can significantly perturb the asteroid's rotation, especially as the scale of material loss grows.

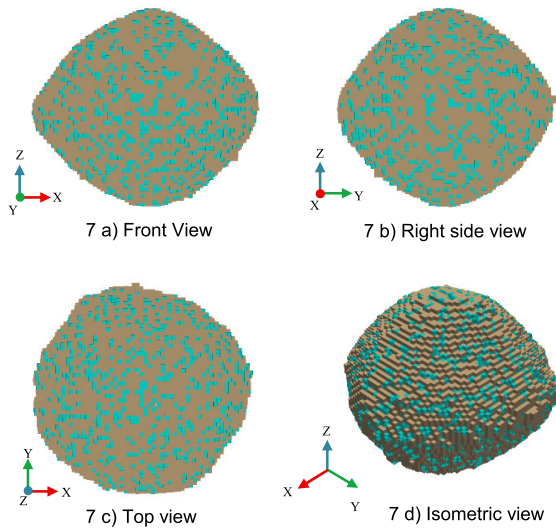


FIGURE 7. Final shape of the asteroid after mass removal in scenario 1.

In both scenarios, the optimized removal strategy produces residuals that are almost flat and centred at zero over the period, both indicating that the dynamical disturbance is effectively negligible. This is an important distinction as it shows that the optimizer was effectively able to choose removal points where the effect on the asteroid's rotational state was minimized. Ultimately, these results underscore that optimized, strategically selected material removal not only preserves the asteroid's dynamical stability but also enables higher resource yields without compromising operational safety or introducing severe uncertainties into the body's

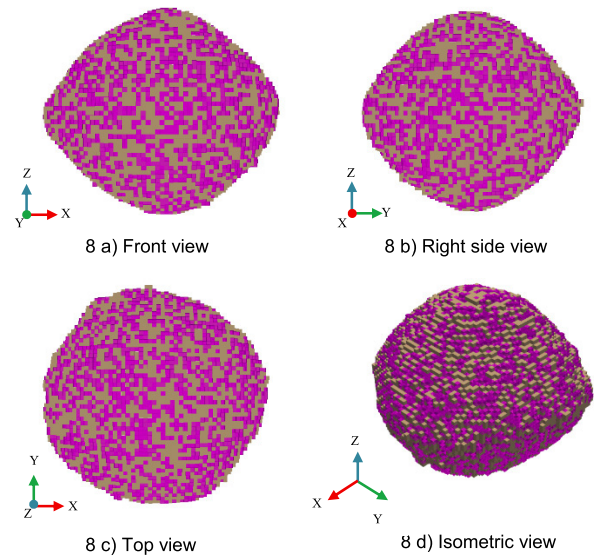


FIGURE 8. Final shape of the asteroid after mass removal in scenario 2.

motion. So, there is no need to install thrusters or any other device to control the rotational instability. This shows the need to develop asteroid mining operational planning systems with the incorporation of optimization capabilities, as well as the risks associated with non-optimized extraction strategies, especially in cases with a higher removal fraction.

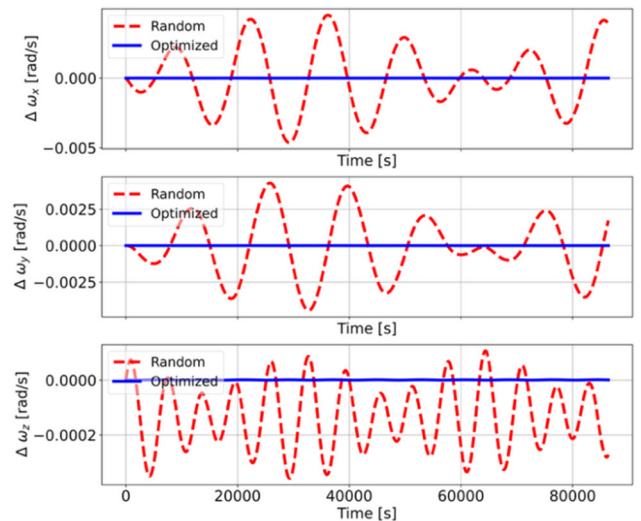


FIGURE 9. Residual angular velocity (the difference between initial and final changes) component over time for 30% material removal, comparing random and optimized extraction strategies.

FIGURE 11 depicts the residual angular velocity norms from both random and optimized extraction of 30% and 70% of the mass, where a clear trend concerning the influence of dynamics introduced by the removing strategy can be seen. For both cases, the optimized extraction had consistently much smaller norms than the random extraction, indicating that the optimizer was able to determine the surface regions where material could be removed with minimal perturbation

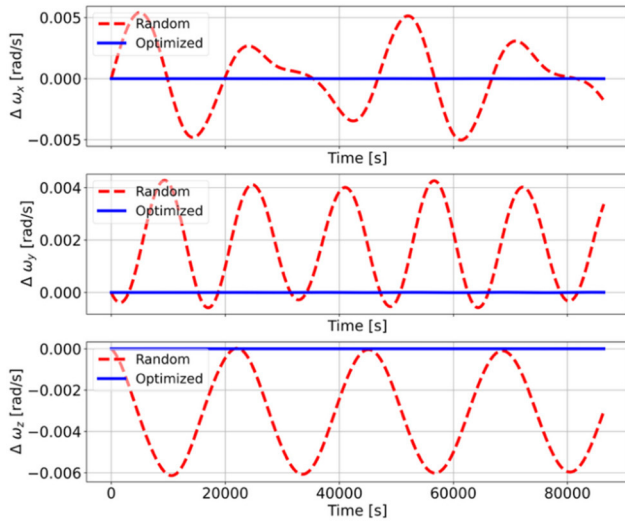


FIGURE 10. Residual angular velocity (the difference between initial and final changes) component over time for 70% material removal, comparing random and optimized extraction strategies.

to the asteroid's rotational state. This advantage is especially pronounced in the 70% removal case, where the random strategy leads to a dramatic increase in residual angular velocity, while the optimized approach keeps the error several orders of magnitude lower.

Additionally, as the fraction of material removed increases from 30% to 70%, the residual norms for random removal increase significantly. Random removal leads to greater dynamical disruption due to material removal being uncoordinated. Optimized removal leads to very low residual norms even at a very high fraction of removal. In other words, optimum removal is both robust and effective. These results highlight the critical importance of strategic, optimization-driven resource extraction for preserving the asteroid's rotational stability and structural integrity and strongly support the adoption of such approaches in future asteroid mining operations.

FIGURE 12 shows the inertia tensor changes over simulation time. Both scenarios are simulated in the same period. The behaviour of the principal moments of inertia (I_{xx} , I_{yy} , I_{zz}) shows a decreasing trend over time for both Scenario 1 and Scenario 2, indicating that the mass distribution of the asteroid changes as material is removed. Scenario 2 (red) illustrates lower moments than Scenario 1 (blue). This suggests that Scenario 2 involves removing mass in a manner that results in a greater reduction in the asteroid's overall inertia along the principal axes, which may reflect a different excavation strategy or target area. Regarding the off-diagonal components (I_{xy} , I_{yx} , I_{yz}), the more complex and scenario-dependent behaviours can be observed. For the off-diagonal values, it can be distinctly shown that both scenarios increase in the disturbance of stability and the inertia tensor. Those differences could arise from the locations of the mass removal, as one of the main goals of mass excavation is to ensure the stability of the asteroid. Those results are

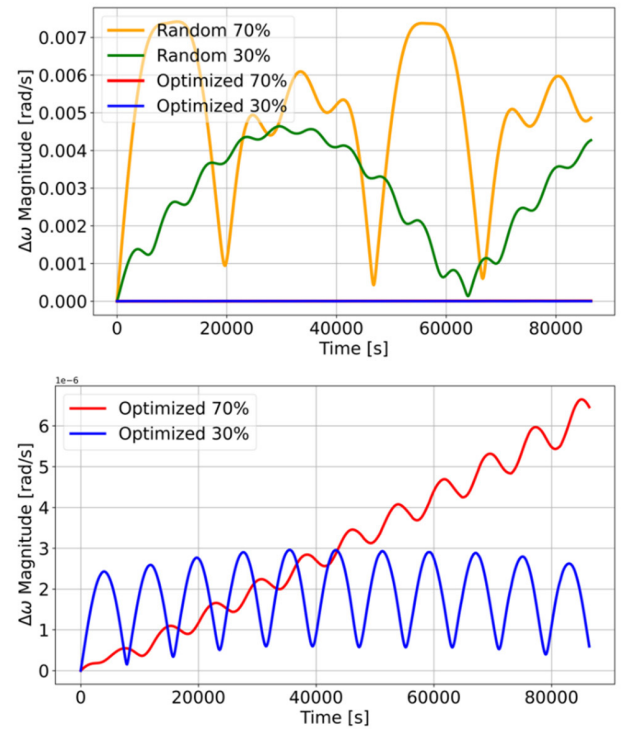


FIGURE 11. Comparison of the magnitude of residual angular velocity (norms) over time for 30% and 70% material removal, using both random and optimized extraction strategies. The bottom figure is a zoomed-in view of the upper one, provided to better compare the two.

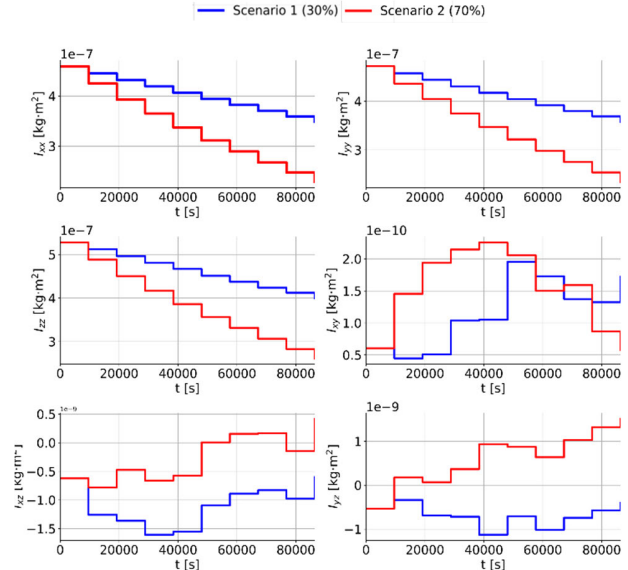


FIGURE 12. Change of inertia tensor for both scenarios over time.

promising signs of the effectiveness of the mass excavation, as the asteroid retains its original characteristics.

FIGURE 13 illustrates the conservation of energy and angular momentum, which is validated for both scenarios. The energy remains almost constant over time, indicating that the system conserves energy in the absence of external forces. Also, these findings confirm the numerical accuracy of the simulation. In summary, the steady state of these parameters

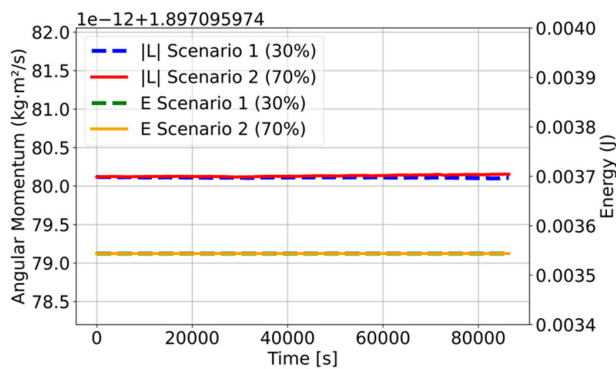


FIGURE 13. Energy and angular momentum changes in the whole period of simulation time for both scenarios.

implies that the system conserves both energy and angular momentum under the simulated conditions.

VI. CONCLUSION

This paper presented a comprehensive investigation into the effects of mineral extraction on the rotational dynamics, with a special emphasis on producing appropriate removal methodologies for asteroid mining. We simulated random and optimization-based material removal for two scenarios where 30% and 70% of the asteroid's surface material were removed to quantitatively investigate the changes in inertia tensor, angular velocity and the overall shape of the asteroid. The results demonstrate that random removal, especially at

higher fractions, leads to significant dynamical disturbances manifested as large residual changes in angular velocity and noticeable deformation of the asteroid's shape. In contrast, the optimization algorithm consistently identified surface regions where material could be extracted with minimal impact on both rotational stability and structural integrity, even when up to 70% of the surface material was removed.

These results emphasize the importance of planned optimization-based planning for asteroid resource extraction missions. While indiscriminate extraction poses a significant threat of risk, including rotational destabilization and other challenges for the mission based on the asteroid's shape, targeted extraction has the potential to yield a high-value, low-risk alternative with minimal impact on the stability of the asteroid. This work provides a physics-based basis to assess the impacts of surface operations on small bodies, as well as heuristic design aids, to enable safe and sustainable development of asteroid mining missions. Also, by implementing an optimized extraction scenario, the need for active control via some devices like thrusters is eliminated, which reduces the complexity of the mining mission and significantly lowers operational costs. This simplification not only makes the mission more economically feasible but also improves overall mission reliability by minimizing the risks associated with controlling asteroid rotational dynamics during material removal. As a result, optimized mining methods would be important for the sustainable and affordable utilization of asteroid resources.



FIGURE 14. Stepwise visualizations of asteroid surface material removal for the 30% removal scenario.



FIGURE 15. Stepwise visualizations of asteroid surface material removal for the 70% removal scenario.

APPENDIX

This appendix includes a sequence of images that track the changing configuration of the material removal process on the surface of an asteroid for both cases studied here (30% and 70% removal fractions, FIGURE 14 and FIGURE 15, respectively) to show the order-based removal methods, from surface to core, as one of our constraints. The visual progression highlights how material is systematically stripped in layers, with removal beginning at the outermost surface and propagating inward toward the core. This approach ensures that any mass loss process follows physical feasibility and avoids unrealistic scenarios.

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